

Sheets

RP2040



File: rp2040.kicad\_sch

FPGA



File: fpga.kicad\_sch

FPGA2



File: fpga2.kicad\_sch

VGA



File: vga.kicad\_sch

Storage



File: storage.kicad\_sch

Interfaces



File: interfaces.kicad\_sch

FPGA

FPGA2

VGA

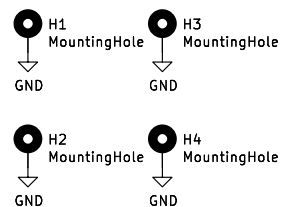
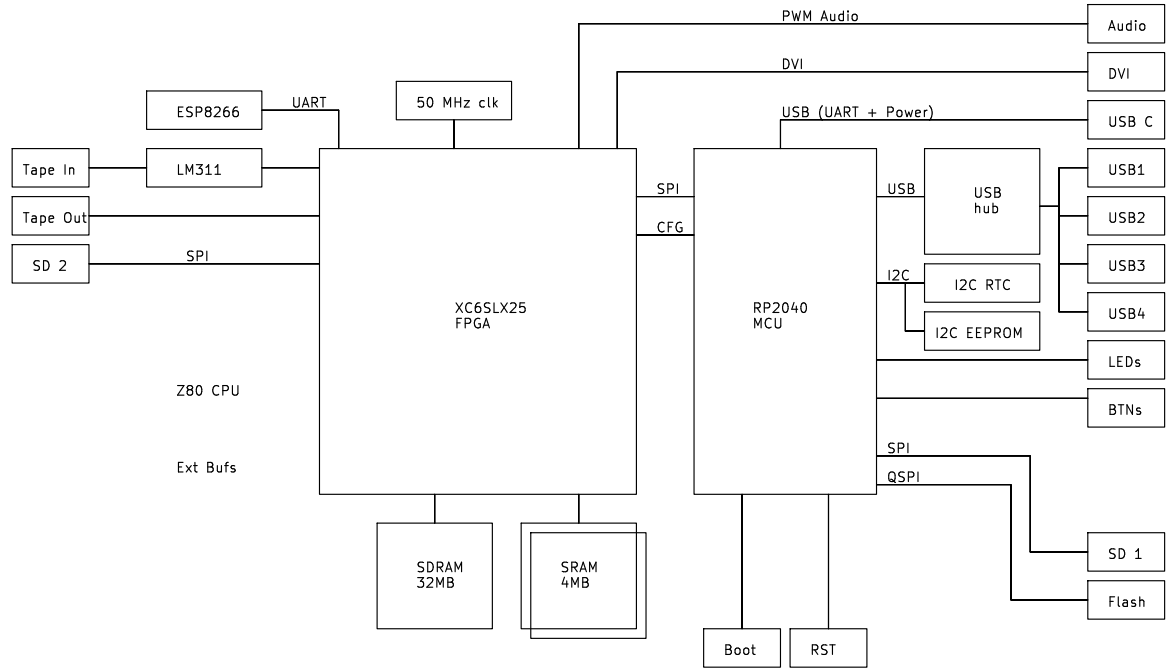
File: vga.kicad\_sch

Storage

File: storage.kicad\_se

#### INTERFACES

The functional schematic illustrates the Z80 CPU system architecture. The central component is the XC6SLX25 FPGA, which interfaces with the RP2040 MCU. The FPGA is connected to a 50 MHz clock, an ESP8266 via UART, an LM311 via Tape In/Out, and an SD card via SPI. It also interfaces with SDRAM (32MB) and SRAM (4MB). The RP2040 MCU is connected to the FPGA via SPI and CFG. It interfaces with various peripherals including Audio, DVI, USB C, a USB hub, I2C RTC, I2C EEPROM, LEDs, BTNs, SD 1, and Flash. The system also includes a Boot button and a RST reset line.

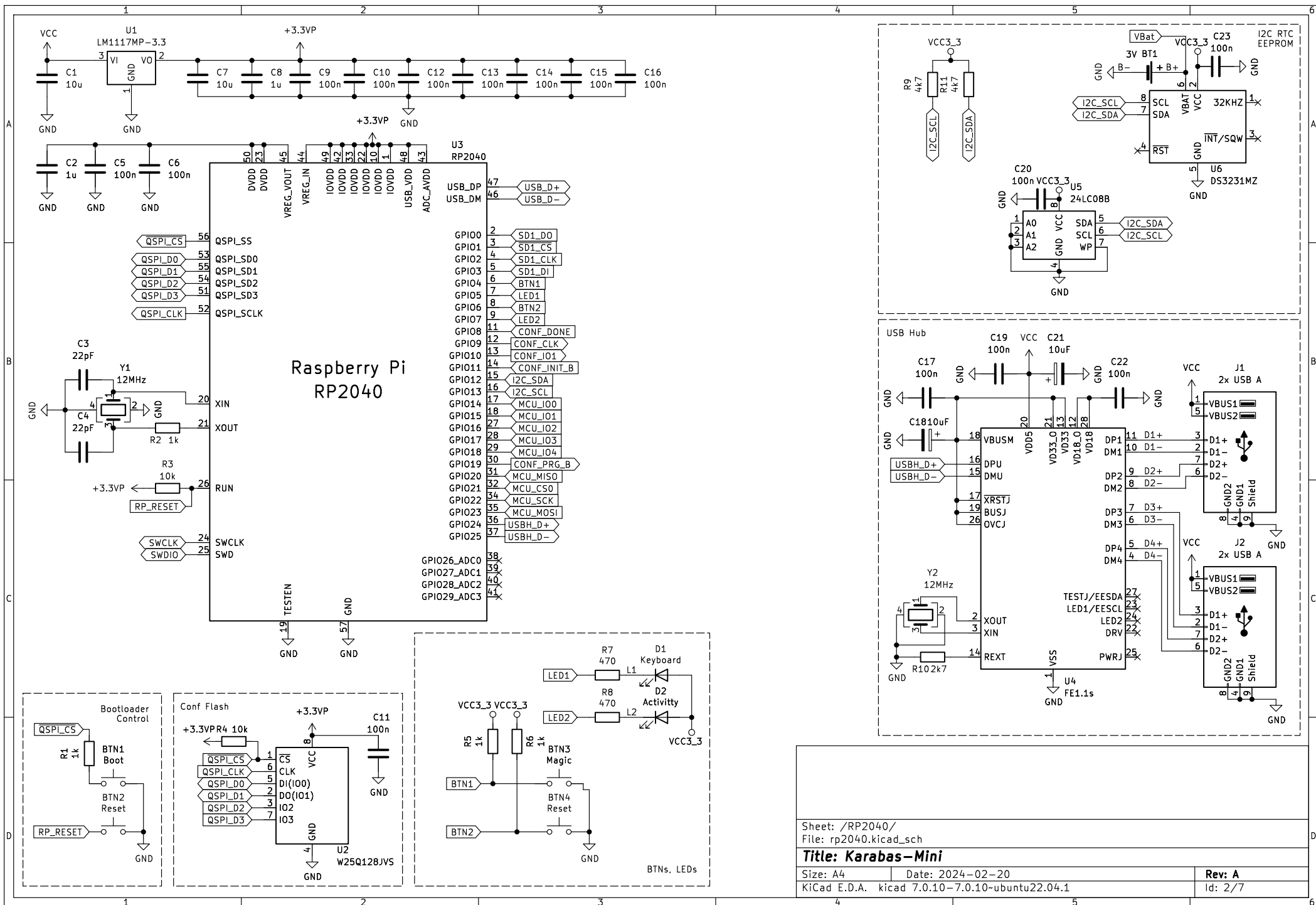


|                                                |                  |         |
|------------------------------------------------|------------------|---------|
| <b>Title: Karabas-Mini</b>                     |                  |         |
| Size: A4                                       | Date: 2024-02-20 | Rev: A  |
| KiCad E.D.A. kicad 7.0.10-7.0.10-ubuntu22.04.1 |                  | Id: 1/7 |

|          |                  |        |
|----------|------------------|--------|
| Size: A4 | Date: 2024-02-20 | Rev: A |
|----------|------------------|--------|

KiCad E.D.A. kicad 7.0.10-7.0.10-ubuntu22.04.1 | Id: 1/7

Id: 1/7



Sheet: /RP2040/  
File: rp2040.kicad\_sch

**Title: Karabas-Mini**

Size: A4 Date: 2024-02-20  
KiCad E.D.A. kicad 7.0.10-7.0.10-ubuntu22.04.1

Rev: A  
Id: 2/7



